# Week 2

**Course Level Objectives:**

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)

| Module Objectives | Instructional Materials | Learning Activities | Assessments | Tools |
|---|---|---|---|---|
| **1.** Distinguish a population from a sample. (CLO 1)<br>**2.** Explain how statistical inference uses sample data to draw conclusions about a population. (CLO 1)<br>**3.** Identify sampling bias and classify the sampling method (random, convenience, or volunteer) in a described study. (CLO 1)<br>**4.** Evaluate whether a study's results can be generalized to the population. (CLO 1)<br>**5.** Recognize bias from question wording, question context, and inaccurate responses. (CLO 1) | e-Text: 1.2 Sampling from a Population<br>Interactive Lecture: 1.2 Sampling from a Population<br>Slides: 1.2 Sampling from a Population (includes sampling video) | Participation: Variables in Statistics (Kahoot!) – Mon: review of Week 1<br>Participation: Sampling – spot the bias in sample scenarios; Your Turn: Name the Bias<br>Participation: Sampling Methods (Kahoot!) – Wed: sampling review | Online Homework: 1.2 Sampling from a Population (due Sun Sep 6, 11:59 PM)<br>Take-Home Quiz 1 (due Wed Sep 9, 11:59 PM) – covers Intro, 1.1–1.3 (all objectives)<br>Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives)<br>Test 1 Study Guide (all objectives)<br>Practice Test 1 (optional; due Thu Oct 1) (all objectives) | WileyPLUS<br>StatKey<br>Blackboard<br>Kahoot!<br>Respondus |
| **6.** Classify a study as observational or experimental. (CLO 1)<br>**7.** Identify a plausible confounding variable in an observational study. (CLO 1)<br>**8.** Explain how random assignment, a control group, and blinding support cause-and-effect conclusions. (CLO 1)<br>**9.** Distinguish random sampling from random assignment. (CLO 1) | e-Text: 1.3 Experiments and Observational Studies<br>Interactive Lecture: 1.3 Experiments and Observational Studies<br>Slides: 1.3 Experiments and Observational Studies (includes TED-Ed placebo video) | Participation: Experiments and Observational Studies – classify studies; confounding; random sampling vs. random assignment; Your Turn: Study Design and Scope of Inference (coffee and depression; vitamin D experiment) | Online Homework: 1.3 Experiments and Observational Studies (due Sun Sep 6, 11:59 PM)<br>Take-Home Quiz 1 (due Wed Sep 9, 11:59 PM) – covers Intro, 1.1–1.3 (all objectives)<br>Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives)<br>Test 1 Study Guide (all objectives)<br>Practice Test 1 (optional; due Thu Oct 1) (all objectives) | WileyPLUS<br>StatKey<br>Blackboard<br>Respondus |